

PIYUSH GUPTA

Backend | Generative AI, LLMs | Python, FastAPI | Noida, India
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Education

Thapar Institute Of Engineering and Technology | 2022 – 2026

B.E. (Bachelor of Engineering) in Computer Engineering | CGPA - 7.95/10.0 | Patiala, Punjab

Kendriya Vidyalaya No.2, Bailey Road | 2019 – 2021

12th Standard: 93% | 10th Standard: 92% | Patna, Bihar

Skills

Languages: Python, C, C++, Rust, SQL, JavaScript

Generative AI: LLMs, Prompt Engineering, AI Agents, Semantic & Vector Search, NLP, Hugging Face Diffusers

ML & Data: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Feature Engineering, Model Evaluation, Statistics

Cloud & Tools: FastAPI, Docker, Git, CI/CD, REST APIs, Streamlit, Playwright

CS Fundamentals: Data Structures and Algorithms, Object Oriented Programming, Database Management System, Operating Systems, Computer Networks, Software Engineering

Projects

AI Native API Orchestration Gateway | *Python, FastAPI, OpenAPI, Docker* | [Demo](#)

- Developed an LLM powered API orchestration platform using FastAPI, OpenAPI ingestion, semantic routing, and vector search to convert natural language queries into validated REST API calls across 15+ microservice endpoints.
- Deployed a schema driven gateway with semantic caching, telemetry tracing, Swagger onboarding, and real time service mesh visualisation, reducing manual API integration effort by 80% and improving latency for repeated requests.

IoT Based Early Seizure Detection | *Python, Deep Learning, IoT* | [GitHub](#)

- Led 5 member team to develop a wearable IoT based EEG monitoring system, achieving 94% signal acquisition accuracy for real time epileptic seizure detection culminating in a published patent.
- Optimised a hybrid ML pipeline combining 1D CNNs with handcrafted feature extraction using wavelet, statistical, spectral features, achieving 99% training and 97% testing accuracy across multiple classifiers (RF, LightGBM, XGBoost).

Generative Imagined Speech Decoding | *Python, PyTorch, Deep Learning, GAN* | [GitHub](#)

- Built a PyTorch 1D CNN pipeline to classify imagined speech from 64-channel EEG (preprocessing, artifact rejection, augmentation, class balancing), improving accuracy 23% over baseline to 81%.

Work Experience

Analyst | Smart Energy Water | July 2026 – Present

- Designed and implemented a Python based REST API automation framework using pytest and FastAPI, running 1,000+ test scenarios across 20+ microservices in parallel, reducing test execution time by 60% and improving release readiness.
- Integrated automation with SQL databases across 10+ microservices, enabling real-time data extraction and API-to-database validation, reducing manual verification by 40%.

Trainee | Smart Energy Water | January 2026 – July 2026

- Built Playwright based UI automation covering 100+ end to end scenarios, integrating step level logging and video capture into a CI/CD pipeline, reducing manual QA effort by 70%.

Product Intern | Samsung Prism | October 2025 – December 2025

- Engineered a ControlNet based image generation pipeline using PyTorch and Hugging Face Diffusers, fine tuning Stable Diffusion v1.5 for controllable 2D image generation from wireframe and skeleton inputs.
- Developed a synthetic data pipeline using Objaverse (800K+ assets), converting 10,000+ 3D meshes into wireframe conditioned training pairs and reducing preprocessing time by 40%.

Publications and Achievements

- Patent** Published on **IoT based Early Seizure Detection System** - (Indian Patent Office, Application No. 202611022883): AI based system for real time EEG seizure detection using IoT edge devices.
- Adobe India Hackathon 2025** - Semi Finalist (Top 2% nationwide): built a GenAI based PDF to snippet & podcast mode system using Azure TTS with an Insight Bulb feature powered by Gemini 2.5 Flash.
- 2nd Runner Up at **HackOwasp 7.0**